

BEYOND NODAL DAMPING

WHEN STABLE REPLACEMENTS FAIL TOGETHER

Minimal Incompatibility Hypergraphs for Safe SG → IBR Transition Planning

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1 MOTIVATION & QUESTION

Replacing SG units changes the full portfolio. A stable endpoint can still contain an unstable intermediate coalition.

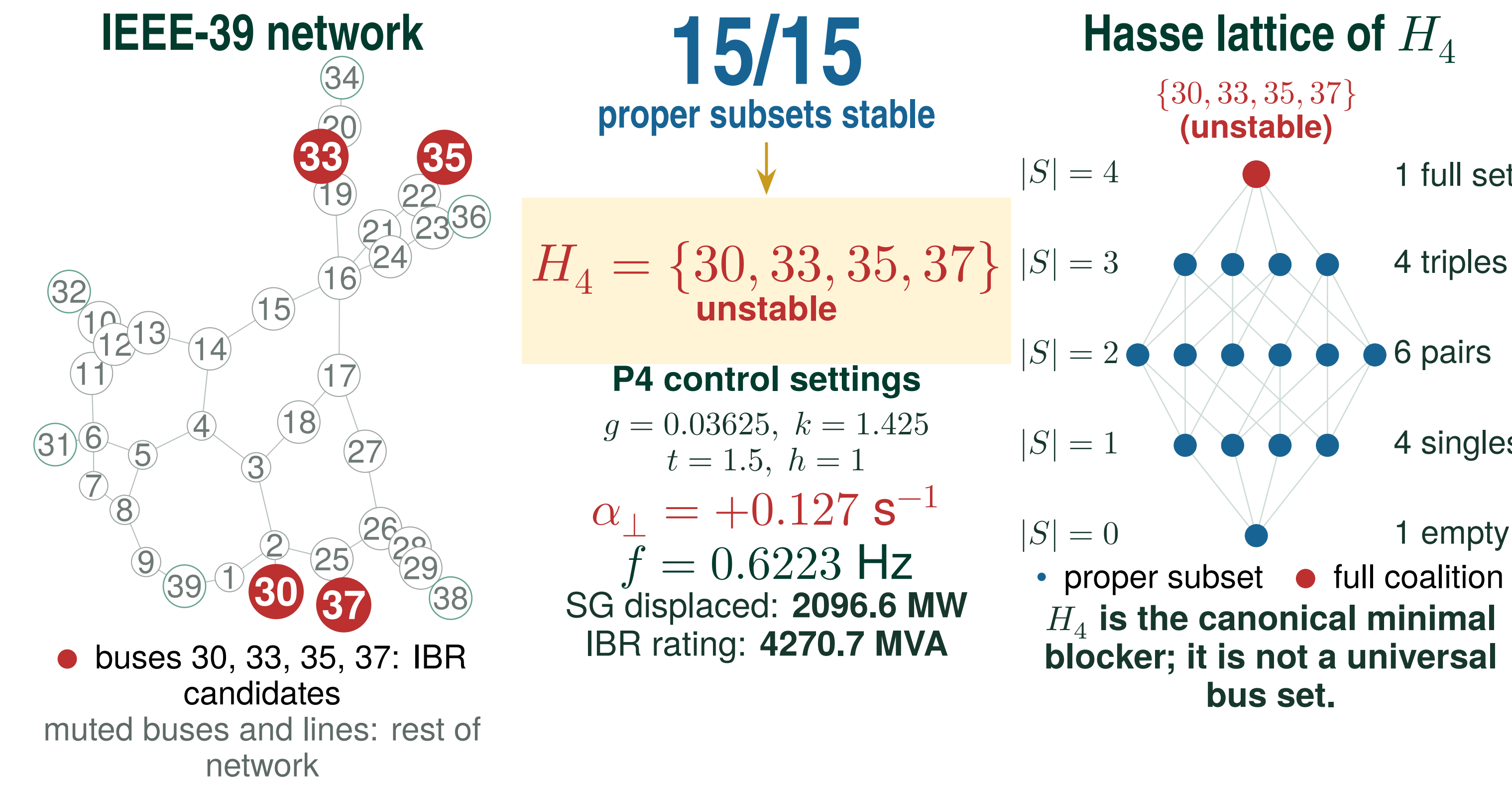


Which replacement combinations are incompatible, why do they fail, and how can paths avoid them?

$$\mathcal{H}(\theta) = \text{Min}_{\subseteq} \{S : \alpha_{\perp}(S, \theta) \geq 0\}$$

Edges mark minimal unstable coalitions.

2 FLAGSHIP RESULT: IEEE-39



3 STABLE TARGET $\neq$ SAFE PATH

Three nested guarantees for a target portfolio T :

A FINAL STATE STABLE $\alpha_{\perp}(T, \theta) < 0$	B SOME ORDER IS SAFE Stable prefix chain exists	C EVERY ORDER SAFE No blocker in T .
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$C \Rightarrow B \Rightarrow A$

Converse implications do not hold.

$$C(T) \iff \forall e \in \mathcal{H}(\theta), e \not\subseteq T$$

Exact $\mathcal{H}$ recovered on the frozen F10 set
Counts of 52 operating points

first-order modal	22/52
additive	31/52
pairwise	38/52

15/15
smaller portfolios stable
all strict subsets

$H_4 = \{30, 33, 35, 37\}$
canonical witness
not the universal blocker

$+0.127 \text{ s}^{-1}$
critical growth rate
canonical portfolio

0.6223 Hz
critical frequency
nominal IEEE-39 case

4 THE DEVICES REMAIN REGULAR; THE NETWORK CLOSES THE INSTABILITY

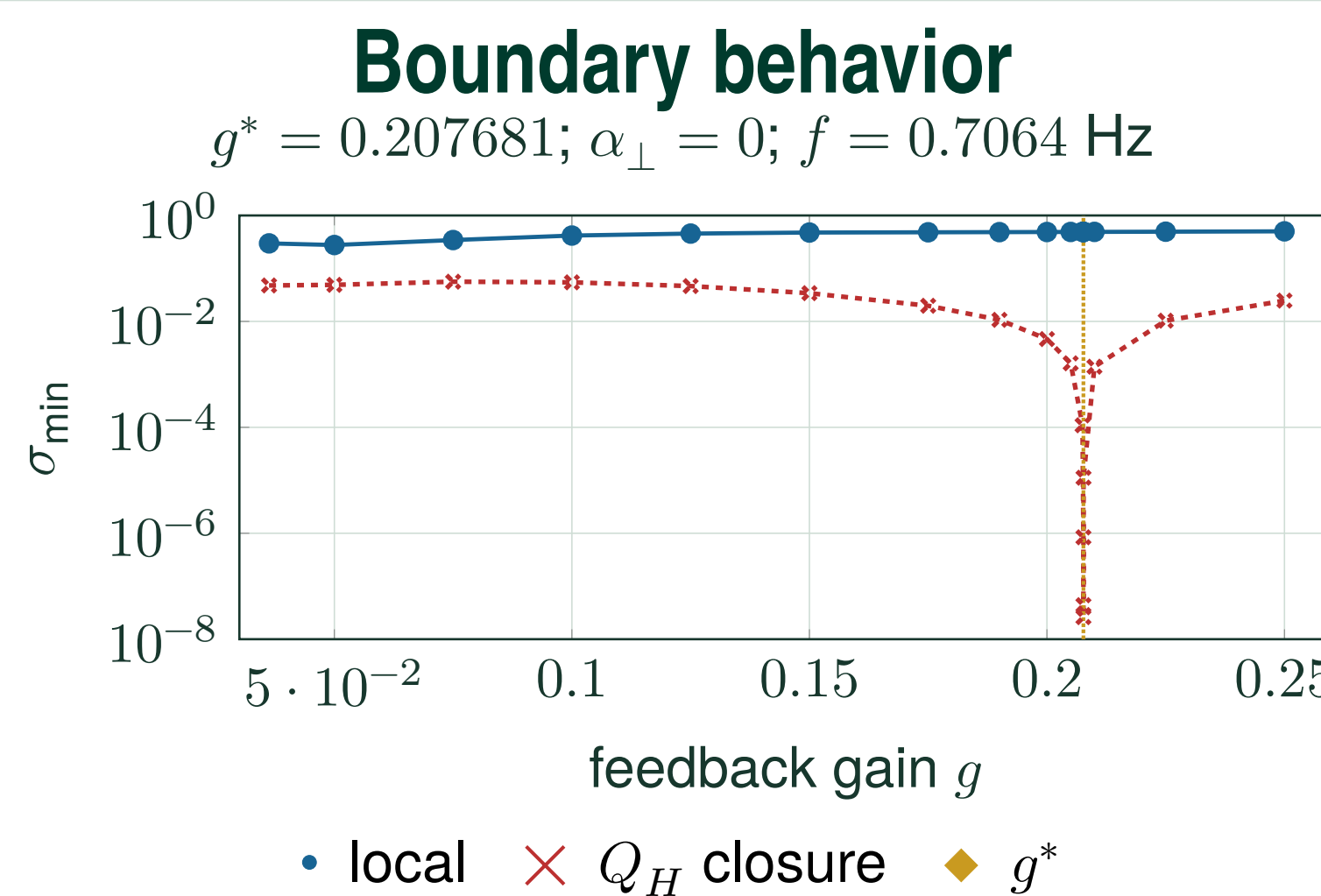
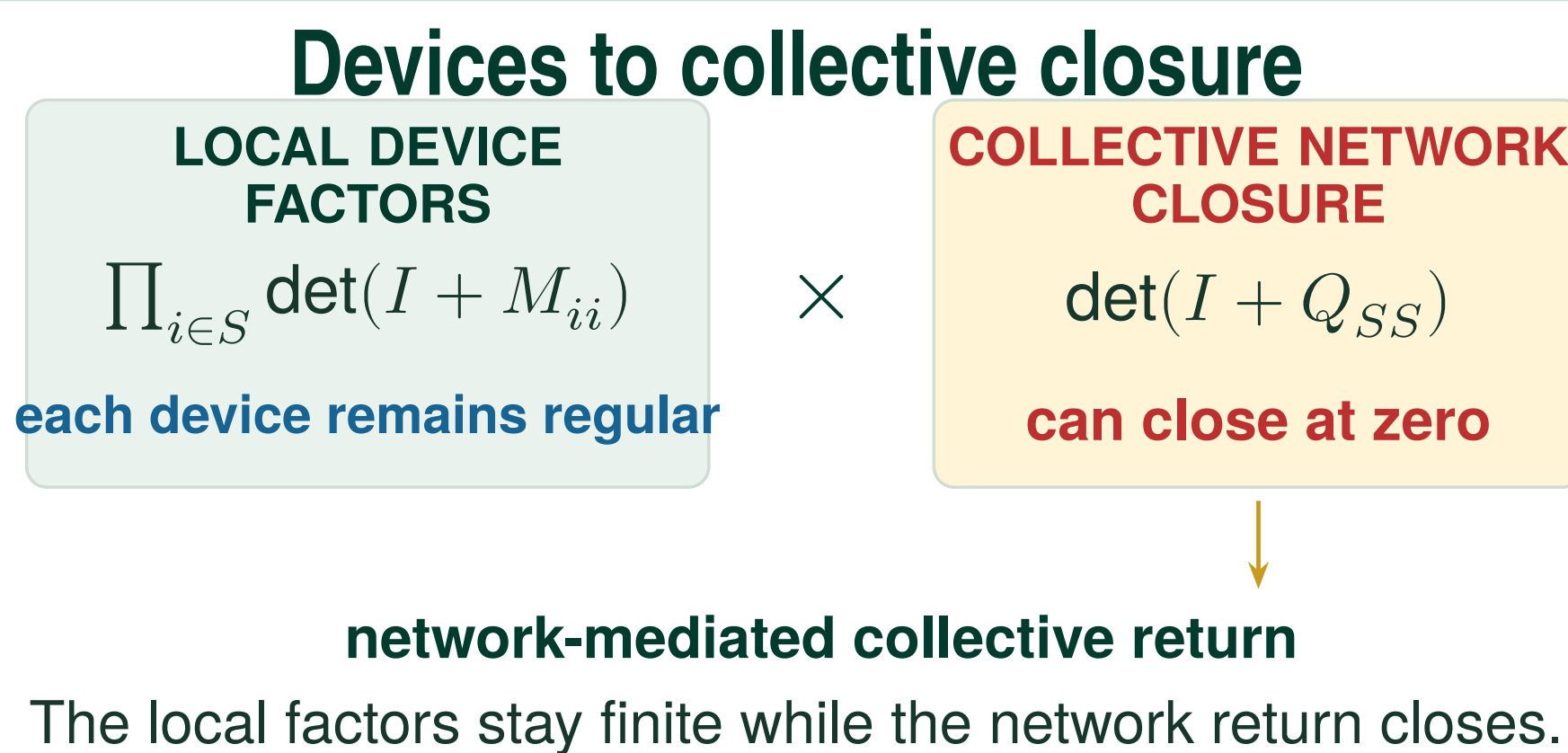
Exact factorization

$$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$$

$$\det P_S(s) = h_S(s) \det T_0(s) \cdot \det(I + M_{SS}(s))$$

$$\det(I + M_{SS}) = \prod_{i \in S} \det(I + M_{ii}) \cdot \det(I + Q_{SS})$$

$$\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$$

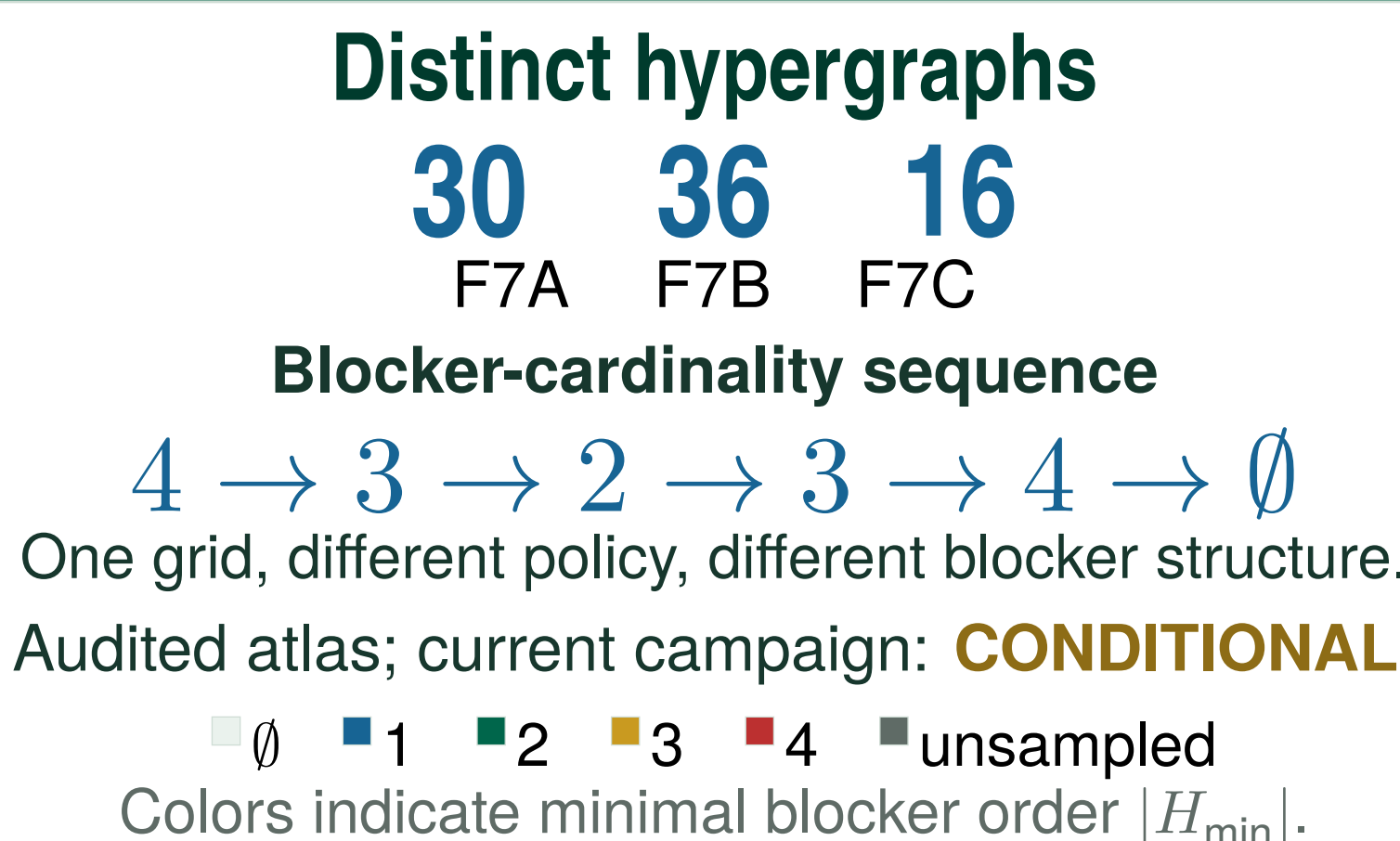
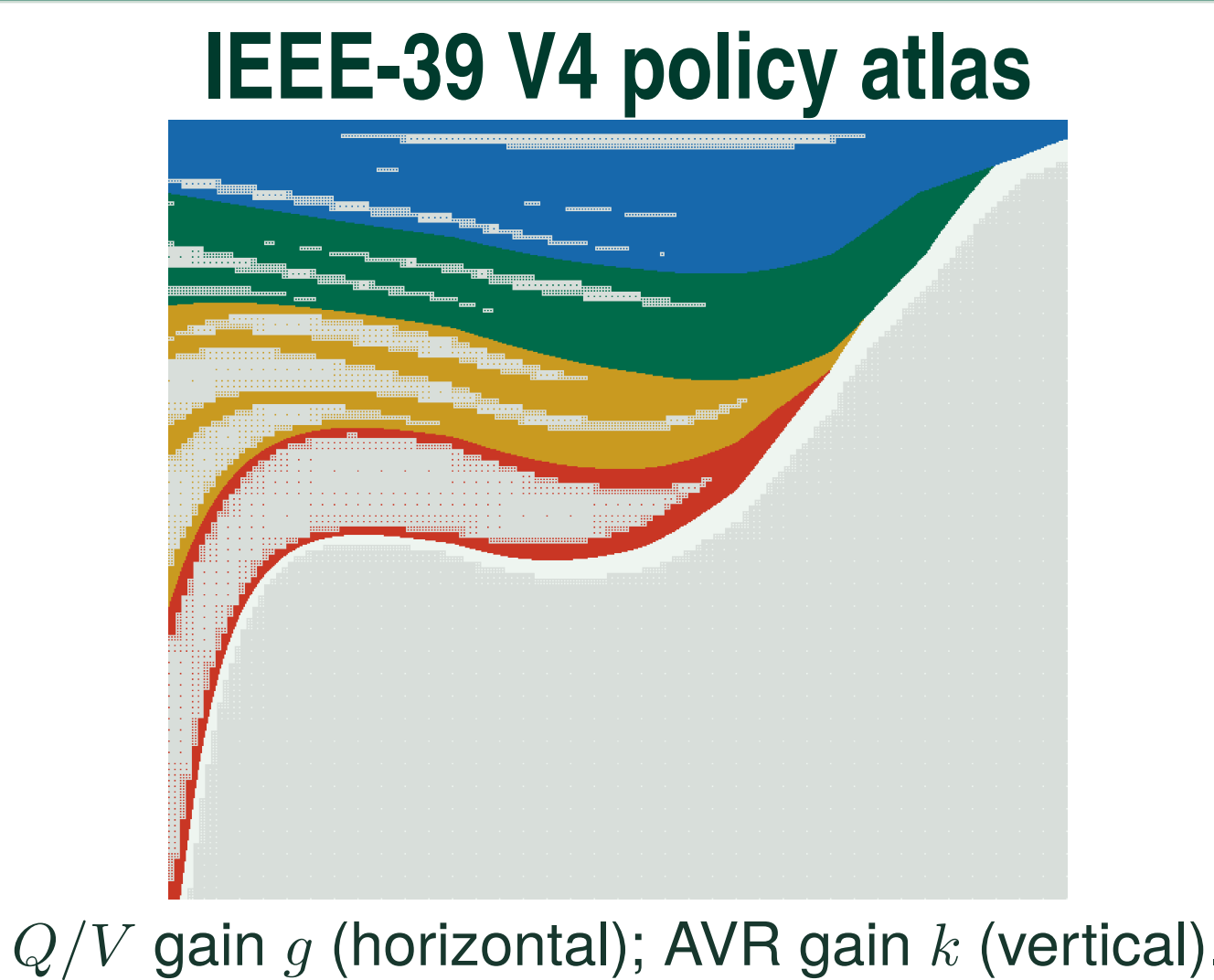


Every member is necessary in context
 H_4 is minimal; all proper subsets remain stable.

The devices remain locally regular.
Instability appears when the network-mediated collective return closes.

Strict M1 reduction: BLOCKED.
 $r = 2.34 \times 10^{-11}$; gate 1.00×10^{-12} .

5 POLICY CHANGES THE MINIMAL BLOCKER



6 THE CANONICAL H_4 IS NOT UNIVERSAL

Frozen synthetic envelope (1000 scenario IDs)



33 infeasible (3.3%) Fractions below use 967 valid IDs.

190/967 19.6% H_4 minimal	555/967 57.4% any V_4 collective blocker	365/967 37.7% alternative witnesses
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SCENARIO TOTALS: PENDING USER REVIEW

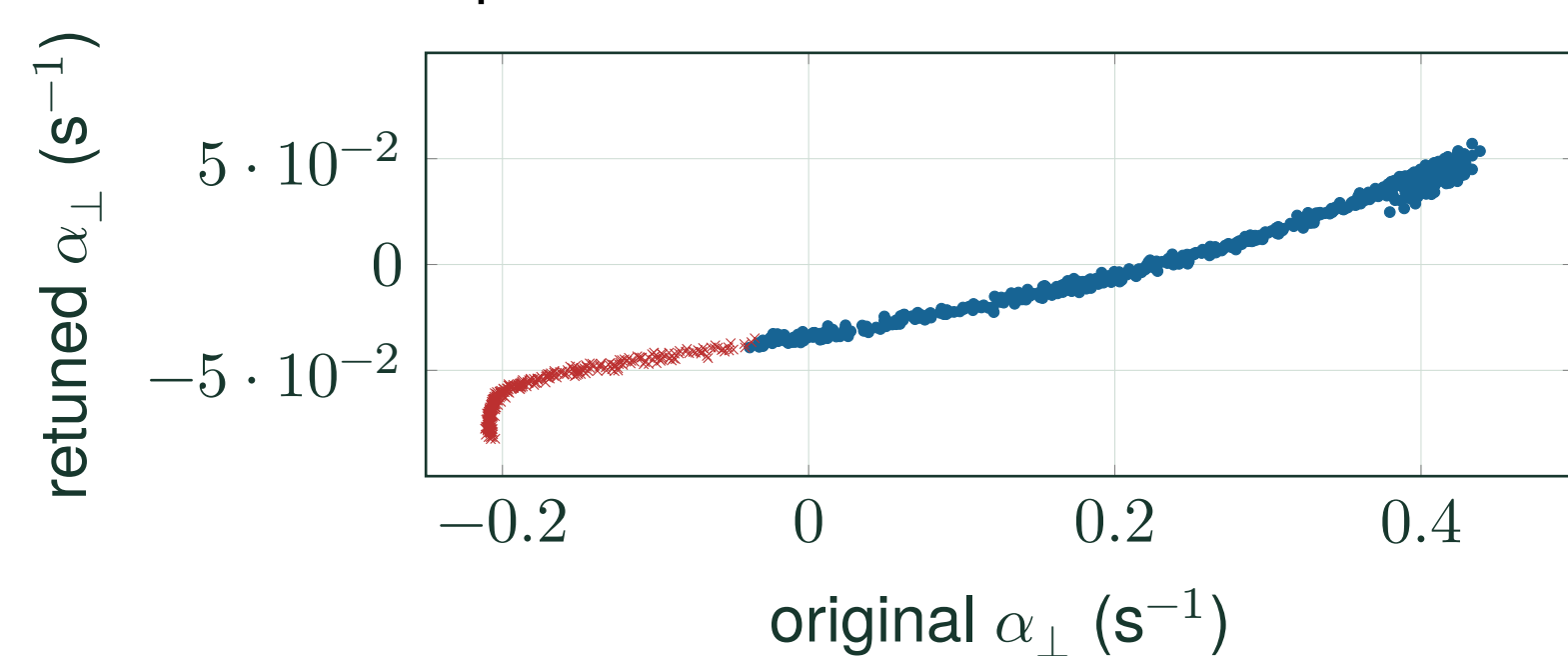
Descriptive synthetic ensemble; IDs are not a probability sample. H_4 is minimal in the audited event.

7 A CANONICAL REPAIR IS NOT UNIVERSAL

Control retuning: $g = 0.03625 \rightarrow 0.25$

594/967 61.4% $\alpha_{\perp}$ improves	186/967 19.2% canonical H_4 rescued	373/967 38.6% stability deteriorates
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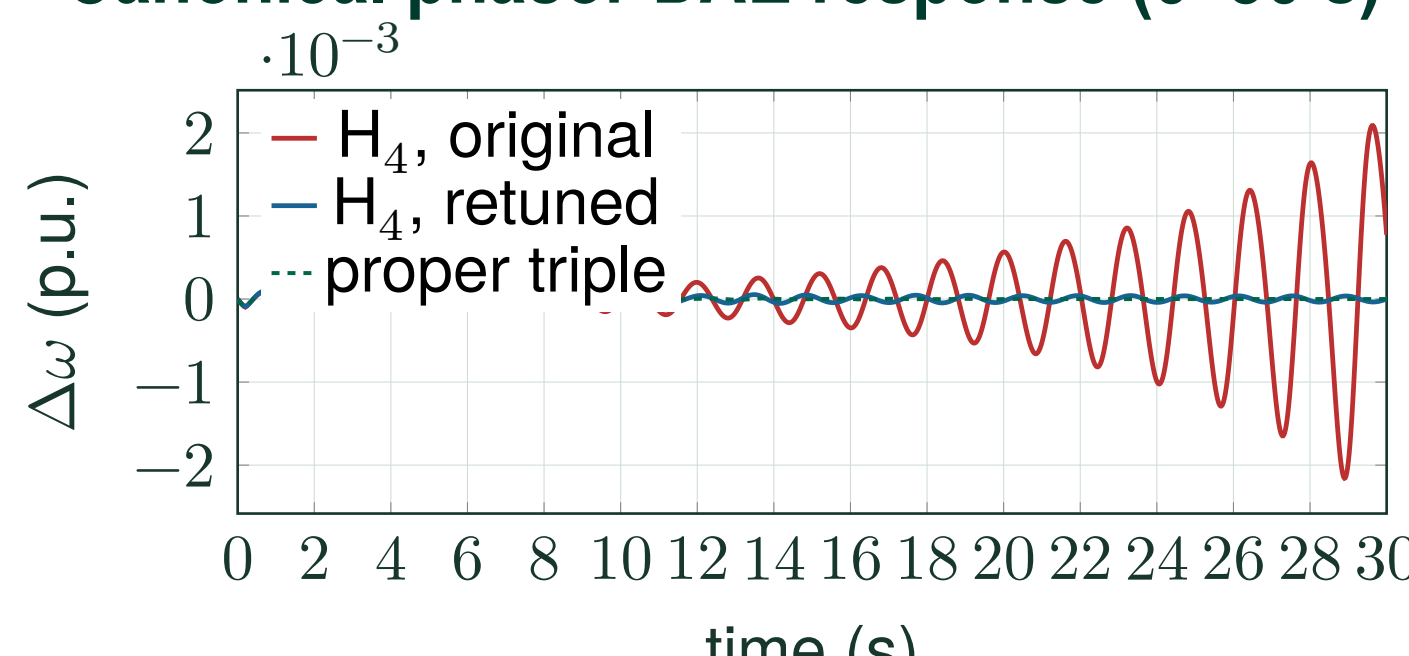
• improved × deteriorated



Support: 106.0 MVA, conditional

Scenario review: PENDING USER REVIEW

Canonical phasor-DAE response (0–30 s)



TDS 56/60; 4 zero-Hz; 30 failed

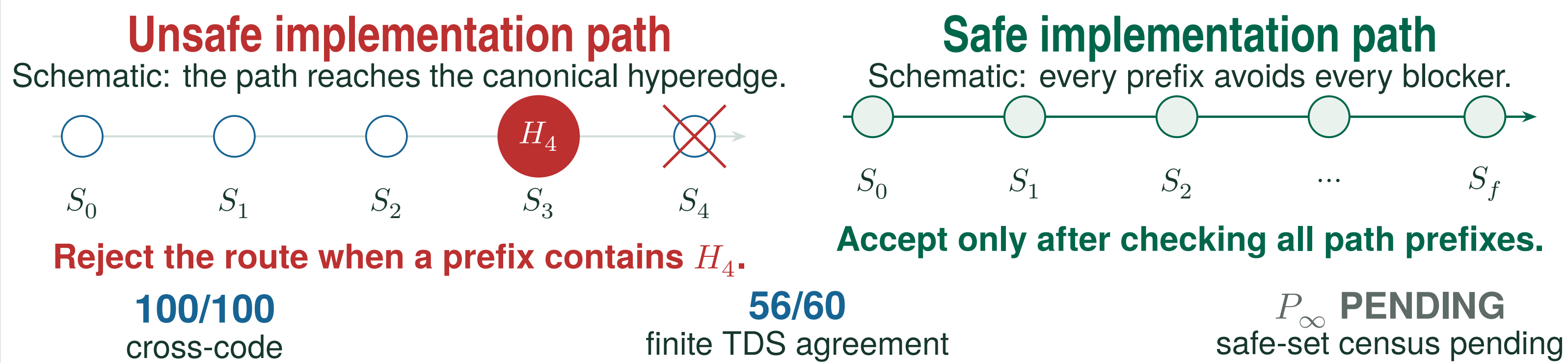
Envelope: PENDING

8 SAFE PATHS AVOID EVERY HYPEREDGE

Path-constrained planning formulation

$$\max_S \sum_{i \in S} P_i \quad \text{s.t. } e \not\subseteq S, \quad \forall e \in \mathcal{H}(\theta)$$

Apply the same constraint to every intermediate prefix of the implementation path.



9 MEASURED EVIDENCE AND SCOPE

100/100 same-model cross-code	56/60 finite TDS sign agreement	15/15 proper subsets stable	30/36/16 distinct policy blockers	594/967 improved; review pending	190/967 H_4 minimal; review pending
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Scope: phasor-domain DAE; stationary, re-equilibrated portfolios; no physical switching transient, EMT, current-limit, or DC-link claim; blocker identity depends on model and control policy.

TAKEAWAY

SAFE TRANSITIONS REQUIRE SAFE PREFIXES.

Check every portfolio prefix against the minimal blockers of its control policy.

REFERENCES

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[5] Milano, *Power System Modelling*, 2010.

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CODE + DATA